Can sonification aid the perception of complex visual networked data? A tool for the interactive exploration of bibliometric data

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Abstract

A side effect of the exponential growth of scientific publications is the generation of large bibliometric databases, dense with metadata. The communication and understanding of these data is complex but can be improved through multimodal representations and interactive exploration. Previous interactive solutions for bibliometric databases rely mainly on visual representations. However, these representations tend to be hard to read, especially when working with large databases. Multisensory perception research suggests that an overloaded visual channel can be supported with auditory cues. Following this principle, we propose an interactive web-based tool that supports visual representation of bibliometrics through an interactive sonification system. The tool combines visual graph representation with three interactive sonification strategies, each designed to highlight different aspects of the publications represented in the graph. This interface allows users to interactively transform bibliometric parameters into sound features that reveal citation patterns and scholarly relationships.

We conducted a think-aloud test with bibliometricians and UX designers to evaluate the tool, and an online perceptual experiment to evaluate our sonification mapping strategies. Results of the think-aloud test show that auditory display can aid and complement bibliometrics visualization. Results of the online experiment show consistent mapping preferences coherent with shared principles of multisensory perception, e.g., large-sized nodes represented by low-pitched sounds, across different interaction modes and speeds. In conclusion, our tool marks a starting point for the use of sonification in the interactive exploration of complex networked databases.

CCS Concepts

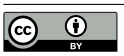
• **Human-centered computing** → *Interaction design*; Empirical studies in HCI; *Visualization*; • **Applied computing** → Digital libraries and archives; Sound and music computing.

Keywords

Multimodal data representation, Multisensory perception, Interactive Sonification, Bibliometric mapping, Data Visualization

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1 Introduction

Scientific publishing grows exponentially [11], and, as a by-product, the size and content of bibliometric databases increase, with vast amounts of metadata on research outputs. Making sense of this large amount of data and using it for research or adjacent purposes requires efficient and creative strategies. Data visualization techniques are often used to gain an understanding of bibliometric databases. Visualization techniques used to explore and gather information from bibliometrics usually focus on graphs, networks, and maps. Tools such as Gephi [1], VOSviewer [31], and Cytoscape [28] are often used to visualize bibliometric data using graph layout algorithms, such as the ForceAtlas2 algorithm in Gephi [17]. The latter method represents the network nodes using a force-based simulation, allowing them to be repelled or attracted, creating interesting visual patterns that enable clustering and enhance visual perception and data malleability. However, networked graph visualizations are often hard to read [23] [27], hindering their usability.

Bibliometrics, like any type of data, can be explored using alternative modalities besides visualization. For example, in a recent study, Bornmann and Haegner [3] generated an audio track from bibliometric data of a prominent researcher by "sonifying" research output. Though primarily illustrative and non-interactive, their work introduced the idea of using sonification to explore bibliographic networks and citation flows.

Interaction is a possible way to help users of complex graphs better understand them [14]. As the authors of the Gephi software highlight, the difficulty of finding features in the visualization of complex networks underscores the need to adopt an exploration strategy [1]. Several studies have introduced interaction strategies with graphs [14, 27]. Previous studies have suggested that the use of multiple and overlapping sensory cues can enhance the comprehension of ambiguous and inaccessible visual information, and impact users' performance and workload [20, 25].

Although visualization techniques have been widely used to display bibliometric information, and there have been some attempts to sonify such data, no attempts have been made to merge visual and sonic modalities into a single multimodal interactive tool and leverage cross-modal perception [32].

In the study presented here, we aim to fill this gap by employing an interactive, multimodal interface for data exploration via force-layout graphs. The interface is applied to bibliometrics exploration and sense-making. We investigate whether adding redundant multisensory mappings to such graphs can enhance perception and

understanding of information. We hypothesize that this addition will not only improve users' data perception but also provide new perspectives, leading to increased information retrieval. We also investigate whether established crossmodal preferences (e.g. low pitch = large size node) hold in interactive graph exploration when the scope is information retrieval. We explore this in comparison to traditional Western music voice leading rules, where the most salient elements are usually positioned in the high-pitched region (high pitch ≃ salience) [16].

For the use case presented in this study, we collected data using a Scopus query, and we used only specific metadata (co-citation coupling, bibliographic coupling, citations). We demonstrate our hypothesis by designing three distinct interaction modes to provide the user with different ways to perceive and interact with the data. The first one is single-node sonification, obtained through node selection, hover, and arrow keys navigation. The second one is a rhythmic visualization/sonification from a selected node. The last one is a transition visualization/sonification that animates the shortest path between two selected nodes.

We evaluated the system twice: first, we studied the interface using a think-aloud method paired with interviews; second, we conducted an online perceptual study to evaluate our sonification mappings. The think-aloud and post-interviews were then analyzed using thematic analysis. The online experiment results were analyzed using chi-square tests, exact binomial regression, and Aligned Rank Transform ANOVAs (ART-ANOVAs). In contrast to earlier studies, which used static sonification, our focus here is on interactivity, making the sonification more closely tied to the data and enriching data perception, confirming findings in previous studies [25].

Results from the think-aloud test show that the multimodal graph interface is well-received by users and enables them to gain new insights from the data. Results from the online experiment show consistent mapping preferences among participants.

The research questions for this research were: (**RQ1**) How does the multimodal graph interface support bibliometric data exploration? (**RQ2**) How do the interaction modes affect the perception of the sonified bibliometric data? (**RQ3**) Do established crossmodal preferences hold in interactive graph exploration when the scope is information retrieval?

The remainder of this paper is structured as follows. Section 2 presents a review of previous research on bibliometric visualization, sonification, and multisensory interfaces. Next, we present the methodological aspects in Section 3. The think-aloud and online listening test experiments and their results are presented in Sections 4 and 5, respectively. In Section 6, the findings are discussed. Finally, we present the concluding remarks and possible future work in Section 7.

2 Related Work

2.1 Bibliometrics

Bibliometrics is the quantitative study of scientific output. Centered around publication volume and citation patterns, it aims to measure the productivity and impact of research entities of varying nature and size, ranging from small groups of researchers to countries taken as a whole. While it has been used for many years to evaluate research, it has also faced criticism for the use of questionable methods and an overreliance on quantitative indicators, prompting calls for a more responsible approach. Initiatives like the DORA [9] and CoARA [5] manifestos have helped disseminate and develop ideas for good practice in research assessment. Current trends favor shifting from purely quantitative metrics to a more qualitative perspective. This includes moving away from misused indicators like the *h*-index and journal impact factor, recognizing the diversity of research outputs beyond the traditional publishing model, and rewarding virtuous practices like open access.

Still, traditional bibliometrics constitutes a useful toolbox with many relevant applications, including mapping out a research area by: identifying strong collaboration patterns; key subareas; influential works; and distinguishing emerging topics by studying dynamic patterns. To this end, one method consists in constructing networks based on relationships like co-publication, keyword co-occurrence, or, as in this study, co-citation. A co-citation coupling occurs when two publications are cited by the same document, with the strength of the coupling determined by the number of shared citing documents. Publication networks can also be constructed based on bibliographic coupling – defined by the fact that two publications share one or more references – or based on direct citations, each method having its own characteristics. Intrinsically, co-citation offers a forward-looking measure of semantic similarities between publications [30].

2.2 Interactive Visual Data Exploration

2.2.1 Networked Graphs. Networked graphs are used in different disciplines and analytical methods. Social Network Analysis provides several examples of interactive exploration of networked data. Perer and Shneiderman [23] outline that interactive exploration of networks is challenging because it is hard to find patterns in large networks, and current systems are visually overwhelming. They introduced a tool for social network analysis called SocialAction that aimed to tackle these issues by allowing the user to filter nodes, find outliers, aggregate networks, and find link types using a matrix view. Shamir and Stolpnik [27] also addressed these issues by developing a tool for multivariate graph exploration that featured interactive mechanisms such as pop-up interaction, topological queries, statistical queries, and contextual queries. While most of the work has been done in trying to improve the visual representation, there has been little interest in supporting the exploration of networked graphs using multisensory stimuli.

2.2.2 Bibliometrics. Software for bibliometric visualization, such as VOSviewer [31] and Gephi [1], support exploration and information retrieval. VOSviewer constructs distance-based maps and interactive graphical representations from bibliometrics. It implements three visualization modes: network, density, and overlay. Visual features of the network can be edited in real time, and it's possible to find specific nodes in the network using text queries. Both nodes and edges are selectable, and a legend on node sizes is provided. Gephi peculiarity is the use of an algorithm for force-directed graph visualization called ForceAtlas 2 [17], which simulates a physical system to spatialize the network; a useful method to avoid visualization clutter, especially in large graph visualizations. Another

notable peculiarity of this software is the possibility of manipulating the data in table format and computing statistics on the analyzed bibliometric corpus. Both software include easy ways to save the analyzed networks as pictures or pdf. Also, they are visual-based with a static and partially interactive graph representation.

2.3 Interactive Sonification

Sound is an excellent way to represent and communicate the unseen and to raise awareness. Sound can convey information about multiple variables simultaneously, making it a multidimensional information carrier. Sonification is the discipline of translating data relations into non-speech sound relations in a systematic and reproducible way [18]. In a recent overview paper, Bornmann [2] has illustrated how data sonification can be used as a possible alternative to data visualization. In a recent study, very relevant in the context of our research, Bornmann and Haegner [3] defined the sonification of bibliometrics as 'metrics sonification': describing it as "the sonification of bibliometric information (measurements, data or results) for their empirical analysis and/or presentation" [3]. They applied this sonification to the publications of a notable researcher by sonifying three bibliometric parameters: publication output, open access publication, and citation impact of publications. They provided the results as an audio track. Their sonification strategy was non-interactive and focused more on affect rather than information retrieval. A recent study by Lindborg et al. [19], where they analyze the Data Sonification Archive[1], shows that around 29% of collected sonifications pertain to the field of visual with complementary sonic representation. Dayé and De Campo [7] suggest that sonification should be complementary to the visual and can transform data exploration into an experience akin to playing a musical instrument.

In the context of our study, we use interactive sonification that focuses on the real-time representation of data through sound [15]. It is particularly useful for large amounts of data where a graphical display may be difficult to interpret due to limited space.

2.4 Multisensory Data Representation

Hogan and Hornecker [13] define multisensory data representation as "a class of data representation that has a clear intent to reveal insight by encoding data in more than one representational modality and requires at least two sensory channels to fully interpret and understand the underlying data". One multisensory principle that is often used for data representation is that of cross-modal correspondences, defined as compatibility effects between stimuli in different sensory modalities [22]. These influence multisensory integration and perception by enhancing the temporal/spatial binding of congruent stimuli. They are known to impact response speed and accuracy, with their effects varying depending on dimension pairings and tasks. Studies related to Virtual Realities have investigated interfaces that leverage multisensory principles for information retrieval and their development [20, 25]. Nagel et al. [20] showed that, in immersive visual data mining with redundant sound-visual mappings, sound can support visual cues when visual information is occluded. Rey et al. [25] showed that the usage of redundant combinations of sensory mappings in an interactive interface task improved performance and workload metrics. However, multisensory perceptual principles are still not applied to interactive networked graph exploration.



Figure 1: The multi-modal graph interface when loaded with a co-citation graph.

3 Methodology

In their practice, bibliometricians make use of interactive interfaces for data exploration. For example, they use network graph visualizations in tools such as VOSviewer [31] and Gephi [1]. Our hypothesis is that including sonification in the interaction with a networked graph representation may enhance the perception of the data and provide a richer understanding. To investigate this hypothesis, we developed an interactive sonification interface for exploring bibliometrics in force-layout network graphs. In this type of graph layout, edges and nodes are represented graphically as visible in Figure 1, where nodes are the data points and edges the connections between them. In our tool, we designed three modes of multimodal interaction with the graph, each built to enhance the perception of the features of the bibliometric network.

In the interactive sonification process, we focused our design on two elements of the graph: the inter-node relationships and the inner feature of each node. For the data, we chose to use Scopus[2], a well-established bibliometric database, to retrieve publication data and citation information via its built-in export function and APIs. Our interface is developed as a web application while the audio synthesis engine is run locally using SuperCollider [33]. The web application communicates with the audio synthesis engine using Open Sound Control (OSC) and a WebSocket. The interface was evaluated twice. First, we ran a think-aloud test [26] to understand how users interact with the interface and to find potential shortcomings or development avenues. Then, we ran an online perceptual test to evaluate the sonification and cross-modal mappings.

3.1 Dataset

We extracted publication metadata from Scopus using a single query to identify documents containing "sonification" or "auditory display" in the title, abstract, or keywords, obtaining a set of 3501

[1] https://sonification.design

[2] https://www.scopus.com/

publications. We removed duplicates and excluded entries from domains where the search terms were used in a context unrelated to our research area (e.g., treatment by ultrasonic waves for "sonification", animal communication for "auditory display"). This resulted in a final set of 2994 publications spanning from year 1952 to 2025. For each, we retrieved all citing documents and references to construct the co-citation and bibliographic coupling networks, along with other metadata fields from Scopus: publication year, citation count, subject areas, and topic clusters. Additionally, we obtained the open access status from the OpenAlex[3] publication database when available.

3.2 Web Application

The web application was developed in HTML, CSS, and JavaScript, using the libraries D3.js[4] and force-graph [5]. The resulting application is displayed in Figure 1. On the top left of the screen, features of the currently selected node are displayed: title, open access status, number of citations, number of coupled elements (related to the bibliographic or co-citation couplings), and year. On the bottom right, instructions for navigating the interface are displayed. On the top right, there is a menu for controlling interface settings. Features of the data were mapped to the sonification model and/or to the graphical representation. In the graphical interface, node size was mapped to a node's degree, which in our bibliometric network represents the number of coupled elements. The node colour was mapped to the open access status, which can be Gold (Gold), Diamond (Light Blue), Green (Green), Unknown (Dark Grey), Closed (Grey), or Hybrid (Purple). Finally, the connection line width was mapped to the coupling value between nodes.

3.3 Sonification Strategy

The sonification mappings and sound design were controlled both by Supercollider and the web application. Our design goal was to provide the users with an interactive interface that could offer different perspectives on the same data. This meant designing interaction with the purpose of highlighting the inner features of nodes as well as the inter-node relationships. To attain this goal, we created different interaction modes, each coupled with a sonification model. The primary design consideration was the mapping of co-citations and bibliographic coupling to an eventful sonification strategy that could effectively communicate this data. To achieve this, we used sound models characterized by a rich and varying spectrum, which can facilitate quick identification of differences between nodes.

For the audio synthesis, we used a resonator bank, since it provided us with both timbral and temporal control of the sound. We mainly use the Supercollider Ugen "DynKlank"[6], which is a bank of frequency resonators that can be used to simulate the resonant modes of an object. We also focused on making the sound aesthetically interesting by following Cunningham et al.'s framework [6].

[3]https://openalex.org/
[4]https://d3js.org/
[5]https://github.com/vasturiano/force-graph
[6]https://doc.sccode.org/Classes/DynKlank.html

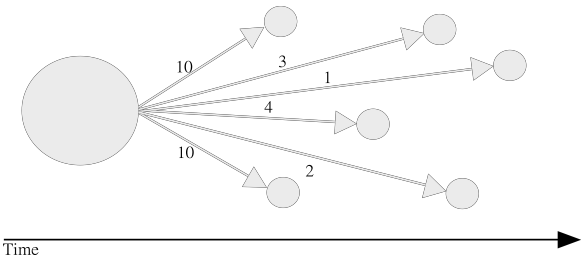


Figure 2: Rhythmic sonification of connected nodes by co-citations/bibliographic coupling. The X axis represents time. The edge's value, representing the coupling, determines the temporal distance from the starting point. The higher the coupling, the closer in time the sounds are produced and vice versa.

3.4 Interaction Modes

The user can interact with the graph using three different interaction modes: (1) single node, (2) rhythmic, and (3) transition. In all these, the user's interactions trigger an audiovisual event.

3.4.1 Single node interaction. The user can interact and inspect the features of single nodes using the following gestures:

- WASD / Arrow Keys - Navigate to the first node in a given direction.
- Node Selection - Triggers when the user clicks on a node; the sound can be retriggered using the "T" key.
- Hover-on-node - If enabled, triggers when the mouse cursor is on top of any node.

When a node is selected, it gets visually highlighted. The sound design focus for this interaction mode is on emphasizing the publication's couplings and citations. In this case, the sonification model does not suggest a relationship between elements, but the user can obtain this information by navigating to other nodes. The sonification is created using the DynKlank resonators filter bank complemented with a sinusoidal oscillator. The frequency and duration of the sound are mapped to the node degree, representing the total number of coupled nodes (high pitch & long duration = high degree), while the timbral quality, in terms of the number of resonators, is mapped to the citation count (high citation = rich timbre).

3.4.2 Rhythmic interaction. This mode of interaction is triggered when a user selects a node while pressing the spacebar. This action starts the playback of all the selected node's coupled elements, with each one of them being played at a different onset time, creating a rhythmic sonification of the node's connections. A visual animation also highlights each node as it gets played. The coupling (i.e. the value representing how many times two papers have been cited/referenced together) determines the temporal distance from the starting point. The stronger the coupling, the closer in time the sounds are produced and vice versa. A graphical representation of this mode is presented in Figure 2, where the horizontal axis represents time. In the visual domain, all nodes are first turned into the gray colour, and, as soon as a node starts playing, it gets highlighted in red colour. In this interaction mode, the focus is shifted from

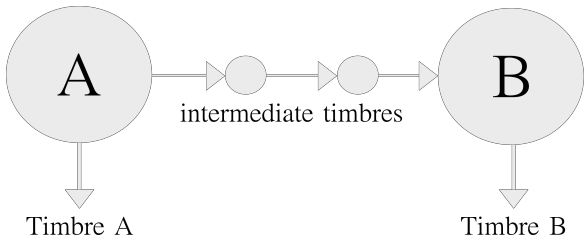


Figure 3: Transition Sonification, the timbre of the produced sound varies during the transition between nodes. The animation starts from a selected node A and moves to a destination node B. The traversed intermediate nodes are also sonified.

the single node to its neighbours. The expected outcome is that of providing a fast indicator of the quantity and quality of coupled nodes.

Two sonification models were created for this interaction mode. The first one is based on a bell-like sound also created using DynKlank, and the other is the single-node sonification model with additional mappings for volume and rhythm. The sound design focus for this interaction mode is on describing the relation and features of node neighbours. The bell-like sound mappings are the degree to the pitch and duration (high degree = big size = high pitch and long duration), and the citations count to the volume (high citation = high volume). The onset time of each sound is calculated from the coupling weight between nodes, with the first node played being the one with the highest coupling value.

3.4.3 Transition sonifications. This mode of interaction is triggered when a user selects two nodes while holding the Shift key. This action causes the shortest path between the two nodes to be highlighted and traversed in time using an audiovisual animation. Since each coupling has a specific value, the proximity in space is defined by the shortest and strongest coupling path, i.e., the path where publications have a higher bibliographic or co-citation coupling.

The sound design focus for this interaction mode is on describing the relationship between nodes through a temporal representation. The sonification is again created using the DynKlank resonators filter bank, but with different mappings. To find the nearest path in the graph, we adopted a *Dijkstra* algorithm [8]. Once the path is found, an animation starts from the first selected node, playing a continuous tone. The timbre of this tone changes as the graph is traversed, sonifying the distinct features (citations and degree) of each stepped node. This tone is generated using the NDef class[7] in Supercollider, which allows for a constant manipulation of the synth with a temporal fade between parameters. With this method, we sought to represent bibliometric features using timbre, as shown in Figure 3. The frequency of the tone is controlled by the node degree (high pitch = high degree), the timbral quality by the citations count (rich timbre = high citations), while the transition time is mapped to the coupling strength between each node pair (high coupling strength = long transition time).

[7] https://doc.sccode.org/Classes/Ndef.htm

4 Experiment 1: Think Aloud

4.1 Study Design

The interface was first evaluated using a think-aloud test [26], with the aim of exploring how people naturally interact with the interface, discovering usability issues, and identifying potential future developments. The experiment was conducted in single sessions with each participant, in the period October 2025 - January 2026.

4.1.1 Participants. A total of five people (4 male, 1 prefer not to disclose) participated in a think-aloud experiment, with an average age of 42.8 (SD 12). Three were professionals in bibliometrics, and two were UX experts. Three participants reported previously using graph visualization for bibliometrics, one participant had used node-based interfaces, and one interactive networked graphs for music creation. The limited number of participants still allowed for identifying usability issues, since it is sufficient to find 85% of usability issues [21]. The participants' identities were anonymized, participation was voluntary and uncompensated. All participants provided informed consent before the experiment began. Breaks were allowed at any time during the experiment.

4.1.2 Equipment. We conducted the experiment in an office setting, with a laptop computer placed on a desk. The laptop's speakers were used for the sound as well as its trackpad and keyboard. We recorded the experiment's audio, the participants' voices, and a video of the laptop's screen. The experience was also manually annotated by a researcher present in the room throughout the experiment. To ensure consistency, all instructions were given in text form, and the researcher did not interact with the participant, unless necessary, until the interview.

4.1.3 Experimental Procedure. Participants were first asked to freely explore the interface while narrating their experience. Then, they were asked to conduct three exploratory tasks on the graph (single node, rhythmic, and transition interaction), each under four conditions (graph type x sound off/on), resulting in 12 tasks per participant. The detailed tasks, as presented to the participants, can be found in the Online Resources 8. The order of the presented graphs was randomized, while they always started without sound. The order of the tasks was not randomized. Participants were all exposed to the different conditions. For each task, the participants were asked to explore the graph while narrating their experience using only one interaction mode. They were explicitly asked to make sense of what they were seeing.

4.1.4 Post interview. After the think-aloud experiment, participants were asked to answer a set of questions in an interview regarding their experience using the interface.

4.1.5 Analysis. The data collected from the experiment and interviews were analyzed using thematic analysis, as proposed by Braun & Clarke [4]. Transcriptions from recordings were created using NoScribe [10].

4.2 Results

The average duration of the experiment was 61 minutes (SD = 12.5). The following themes were generated from the thematic analysis. (1)*Deciphering the Bibliometric Multimodal System* - How

users learned to read and operate the interface, interpreting visual elements, sound mappings, and their relationships. (2) *Affective and Evaluative Responses* - Emotional engagement with the tool: what sparked interest or frustration. (3) *Expectation Alignment and Misalignment* - Instances where interface behavior matched or violated user predictions; sources of confusion and clarity. (4) *Envisioning Use and Improvement* - User-identified applications, missing features, and connections to existing workflows.

4.2.1 Deciphering the Bibliometric Multimodal System. This theme is about how users learned to read and operate the interface, interpreting visual elements, sound mappings, and their relationships. Participants reported they were generally more able to make sense of the data after the introduction of sound. In this regard, participant A5 said, about the rhythmic sonification "*I feel like I'm seeing the ones in the periphery a lot clearer because I'm hearing at the same time. (...) I feel like it tells way more of a story than the visual alone did. With only the visual, I felt like I had no idea really what I was supposed to do. It intuitively didn't really tell me much*". Participant A1 also experienced moments of increased understanding when sound was introduced as an additional sensory aid, helping them perceive differences and make sense of the experiment beyond visual cues.

The meaning of the sonification was not immediate for all participants, for example, participant A3 said "*Not only did it light up the ones that are related to this one, but it also played... Sounds! I'll do that again because I wasn't quite ready for it. Okay, the sounds were... different, but hard for me to tell what they might mean... nothing that directly comes to mind*".

Users initially struggled to understand the nature of the data, but after being presented with the graph with no sound for the second time, participants converged on recognising coupling and node degree over citation as the factor influencing node sizes.

4.2.2 Affective and Evaluative Responses. This theme represents points when the interface sparked curiosity, as well as users' visceral comments on the interface. Participants generally appreciated the visual aspects of the interface, for example, participant A4 noted:"*...this seems so fluid and jelly-like. It's like its nodes are in the water and moving things around...*". Participant A5 also commented on the interface as looking "*underwater*".

Participant A5 also noted how the rhythmic sonification gave them an emotional response: "*That one is so dramatic. Let me do that again. Yeah, that's cool. That's cool. Cause with just the visuals, I got nothing from that really. Like, I don't want to say like emotionally, cause I don't feel, I guess I did. I like, I laughed at that and I liked it, and I said it was dramatic, but it definitely didn't communicate anything intuitively just visually. But with that (sound), it really did*". Both participants A2 and A5 described the transition interaction as a journey. "*It becomes a bit like a movie. A journey through space finding its path in the galaxy of knowledge*" (A2). While A3's comment on the transition sonification was: "*Yeah, these sounds when traveling a path like this is... it's a bit spooky, and I kind of like it.*"

4.2.3 Expectation Alignment and Misalignment. This theme encompasses moments where interface behavior matched or violated user predictions; sources of confusion and clarity. After some time interacting with the interface, participant A1 started to notice patterns and expectations from the interface that later sparked curiosity "*I did it twice in the opposite direction. To see if the part that seemed longer to me actually had a thicker mark. And indeed, that's the case. I mean, it starts out thinner and then gets thicker. Now I'll do it in reverse. So. At this point... I'm curious*".

Throughout the experiment, there were several moments of confusion and misalignment for different participants. One of the most common was related to the interaction with the transition mode. For example, A5 noted "*I accidentally shift-clicked out here somewhere and it immediately lost that other selection. ... I think it should value the fact that I have clicked this and that I'm holding shift. So it probably should know that I want to select something extra. So if I click out, I feel like it shouldn't lose this selection*". Confusion was also common when trying to understand the origin of the presented data. Participant A4 said "*So I'm confused about this too. Like is it, if this is the main paper, this other smaller nodes, are these the references in this main paper? Or are these the papers that have cited this main paper? "*. Other participants noted misalignments when using WASD navigation (expected to move to the closest connected node, not to the first node in the selected direction). The lack of additional visually accessible metadata was also noted by A1, A2, and A3.

4.2.4 Envisioning Use and Improvement. This theme describes users' requests for improvements in the interface. The feature request changed from participant to participant, depending on their intended use for the tool. Their suggestions ranged from increasing information access (legend, metadata) to introducing new interaction modes (edges, clustering) to refining the sound models.

Participants A2 and A3 advised on the introduction of more ways of gathering information from the interface through interaction. Participant A2 suggested the use of a legend: "*I think for me to be able to use this, I would need, like (...) a legend for what different sounds represent,(...) for what different nodes, links, and their thickness, or the distance, how that's calculated (...)*". Participant A4 advised the addition of pop-ups indicating the user's possible error messages.

Participant A1 and A2 also suggested adding the possibility to interact with edges, but advised to avoid over-cluttering the interface "*(...) it should be, maybe, something like an extra feature that, if you press, say, the letter Z and click on the link, maybe then you could find out more about the link. Or some kind of bit hard, like, you know, so you don't destroy the user experience by having too much clutter (...)*". Participant A1 envisioned the possibility of letting users merge nodes into clusters and create a composite sound model.

Participant A5 suggested improvements for the sound model during the transition sonification by adding an additional percussive sound when moving to a new node.

5 Experiment 2: Listening Test

5.1 Study Design

A listening test was conducted using lab.js [12] and hosted on Open Lab [29]. We aimed to evaluate the sonification mappings and cross-modal correspondences (RQ3) across our three interaction modes. To investigate this, we ran a within-subject four-alternative forced-choice (4AFC) where participants were required to match one out of four sonification mappings (pitch/duration or pitch/timbre) to a

visual representation (i.e., a node or visual animation in the graph-layout). The same data was used to generate both the sounds and the visual representations. Findings from Experiment 1 were used to refine the rhythmic sonification model in the listening test.

5.1.1 Hypothesis. We hypothesized that in information-dense conditions, participants would prefer inverted pitch mappings (high pitch = higher importance) to standard pitch-size correspondence (high pitch = small size) [22]. This hypothesis was based on Western music's traditional voice-leading rules [16], in which the leading voice is the one with the higher pitch and the most rhythmic freedom, while the lower pitches are more constrained and less salient.

5.1.2 Participants. A total of 20 participants (10 female and 10 male) participated in the online experiment. Their average age was 33.9 (SD = 9.9). Three participants had no previous musical experience. The recruitment process was conducted online via email and university communication channels. We requested that only participants with self-reported normal hearing could participate. As in the previous experiment, participation was voluntary and uncompensated. Breaks were allowed throughout the experiment.

5.1.3 Stimuli. Participants were presented with 27 visual stimuli, organized in four scenarios. Participants were asked to associate one out of four sonification mappings to each of the stimuli. Sonification parameters used were pitch and duration for scenarios 1 to 3, and pitch and timbral quality for scenario 4.

(1) *Single Node Scenario* - Seven stimuli, each at a different node size corresponding to various degrees (1,5,10,20,30,40,55). The sizes were chosen to sample the full range of degree values in our dataset (min = 1, max = 55). We added more definition to the lower range because it had more data points.
(2) *Navigation Between Single Nodes* - Six stimuli paired with a navigation of different lengths (short, mid, long; i.e. 4,8 and 12 nodes) and speeds (slow, fast; i.e. 1.0 and 0.2 nodes/sec).
(3) *Rhythmic Scenario* - Eight stimuli with rhythmic animation of different densities (low, mid, mid-high, high; i.e. 2, 8, 27, 55 connected nodes) and two speeds (slow, fast; applied to the temporal scaling of the animation, the duration is context dependent).
(4) *Transition Scenario* - Six stimuli with transition animation of different lengths (short, mid, long; i.e. paths of 1,3-4, and 5-7 nodes) and two different graphs (co-citation and bibliographic coupling). Since animations were longer in this interaction mode, participants were allowed to scroll through the video to compare the different stimuli.

For scenarios 1 to 3, the Standard pitch and duration were intended as high degree equals a big node with a low pitch and long duration. In scenario 4, the Standard pitch was intended as high degree equals a big node with low pitch; and Standard timbre as high citations equals a rich timbre. The stimulus duration ranged from 3 to 52 seconds, with an average of 12.13 seconds (SD = 10.19 seconds). The order of Scenarios was randomized, as well as that of the stimuli within each Scenario. Taking into account the feedback from the previous think-aloud experiment, the sound design of the nodes in the rhythmic sonification was updated to follow the same strategy as that for the single-node sonification.

5.1.4 Equipment. Stimuli were presented using the Open Lab online platform. A link to the experiment was sent via email to the participants, who were required to use their computers and a web browser to participate in the experiment. Participants were asked to use headphones.

5.1.5 Experimental Procedure. Participants were first introduced to the experiment's terms and conditions. We then collected demographic data with a brief survey. Afterwards, they were presented with an audio volume calibration step. The participants were then instructed about the experiment's tasks. They were asked to choose the sound option that feels most natural or best fits the visual pattern. Then a familiarization task was presented, which could be stopped whenever participants felt ready. The stimuli were presented in a randomized order within each interaction mode. Participants had to match one of the four sonification modes for each presented animation. After completing the experiment, we ran a survey to collect feedback on the experience.

5.1.6 Data Analysis. The experiment's data were evaluated using statistical analysis in R [24].

5.2 Results

Scenario	N	Std P%	Inv P%	Std D%	Inv D%	Both Std%
Single Node	140	84.3	15.7	71.4	28.6	62.1
Navigation	120	83.3	16.7	70.0	30.0	59.2
Rhythmic	160	76.2	23.8	59.4	40.6	46.9

Table 1: Mapping preferences across pitch (P) and duration (D) (standard/inverted) in different scenarios.

Scenario	N	Std P%	Inv P%	Std T%	Inv T%	Both Std%
Transition	120	79.2	20.8	49.2	50.8	40

Table 2: Mapping preferences across pitch (P) and timbre (T) (standard/inverted) in the transition scenario.

The online test lasted on average 58 minutes (SD=35). Results indicate that, across all interaction modes, standard cross-modal correspondence mappings for pitch and duration are generally preferred. Response distributions across scenarios are shown in Tables 1 and 2. While an a priori power analysis was not conducted, the reported effect sizes (w, h, η^2) indicate that the study was adequately powered to carry on the analyses.

5.2.1 Evaluation of single node, navigation and rhythmic scenario. Responses among the four-alternative forced-choice (4AFC) (standard/standard, standard/inverted, inverted/standard, inverted/inverted), for the navigation, rhythmic, and single node scenarios combined, significantly differed from chance, as indicated by a chi-square test ($\chi^2_{(df=3,N=420)} = 237.24$, $p < .001$, $w = 0.75$).

The pitch (standard/inverted) and duration (standard/inverted) mappings were then analyzed separately using exact binomial tests (chance level=50%). For the pitch mapping, standard mapping was selected for 80.9% of trials, exceeding chance ($n = 420$, $p < .001$, $h = 0.66$). Similarly, for the duration, standard mapping was selected for 66.4%, exceeding chance ($n = 420$, $p < .001$, $h = 0.32$). Mapping preferences were consistent across all scenarios, indicating that the hypothesis based on voice-leading principles was not supported.

5.2.2 Effects of Interaction Parameters. To investigate the presence of interaction effects between parameters, we conducted Aligned Ranked Transform ANOVAs (ART-ANOVAs). In the single-node scenario, a significant main effect of node size on pitch mapping was found ($F(6, 114) = 13.63, p < .001, \eta^2 = 0.42$). Post-hoc pairwise comparisons using Tukey's HSD adjustment showed that Size 5 had significantly higher inverted mapping scores than all other sizes ($p < .001$ in most cases). Size 10 also showed significantly higher scores than the larger size categories (30–55, $p = .02$). No significant differences were found between the larger sizes of 20, 30, 40, and 55. In the rhythmic scenario, an ART ANOVA revealed a significant main effect of length on pitch mapping ($F(3, 133) = 4.27, p = .006, \eta^2 = 0.09$). Post-hoc pairwise comparisons using Tukey's HSD adjustment showed that only the difference between length 10 and 30 was significant ($p < .005$).

5.2.3 Evaluation of Transition Scenario. The transition scenario was analyzed separately because it used a different sonification strategy, focusing on pitch and timbre rather than pitch and duration. A chi-square test indicated that choice among the four mappings differed from chance ($\chi^2_{(df=3,N=120)} = 41, p < .001, w = 0.58$). An exact binomial test on pitch mapping showed that participants' selection exceeded chance ($n = 120, p < .001, h = 0.62$). However, no overall preference for the timbre mapping was found. An ART ANOVA revealed a significant interaction effect between path length and timbre preference ($F(2, 98) = 6.14, p < .003, \eta^2 = 0.11$). Post-hoc comparisons revealed a significant difference in timbre preferences between "Medium" to "Medium Long"($p < .012$) and "Medium Long" and "Short"($p < .006$).

6 Discussion

The study aimed to develop a multimodal tool for bibliometric data exploration that leverages sonification with redundant mappings (i.e., the data represented visually is also the one that is sonified).

(**RQ1**) Our findings from the think-aloud experiment show that the interface was generally positively received. (**RQ2**) Participants of this test also displayed different preferences for the multimodal interface interaction modes, mostly oriented by the information they could retain from them. (**RQ3**) Through the online experiment, we found that mapping preferences are consistent across the different interaction modes of the interface, displaying a strong preference for established cross-modal correspondences.

The think-aloud of the interface yielded various results. Studying users' exploration provided us with insights into how they construct mental models around the interface's elements. After conducting a thematic analysis, four themes were generated: *Deciphering the Bibliometric Multimodal System*, *Affective and Evaluative Responses*, *Expectation Alignment and Misalignment*, *Envisioning Use and Improvement*. In the first theme, we identified that the mental model construction was not immediate, but findings show that information retrieval was enhanced after the introduction of sound. Our data also shows that users' prior knowledge, especially their domain expertise, strongly influenced their sense-making and interface use. Preferences for interaction modes were also influenced by users' scope, and the mode that provided the most relevant/useful information was preferred. We interpret the findings from theme 2 as a general appreciation for the interface. However, certain interaction mechanisms led users to frustration, displaying a need for improvement. The multimodal representation led users to display affect towards the represented data. For example, participant (A5) noted how they were able to memorize and sing the melody of a paper cluster. On theme 3, we interpret the findings as a general necessity for additional visual information in the interface, while maintaining a balance to avoid visual clutter. Given the exploratory nature of the experiment, we didn't provide participants with too much information, but we did try to understand how they constructed mental models of it. This lack of information sparked confusion in participants and hindered their ability to evaluate the usability of the tool. Theme 4 represented the valuable users' comments. The participants' domain expertise made future development directions clear.

The online test showed consistent results: participants significantly preferred standard cross-modal mappings over inverted ones. Pitch mapping preference is consistent, with duration mapping showing a weaker but similar trend. For the single node sonification, we found a significant difference in mapping preferences at small sizes. One possible explanation for this result is that the video stimuli for intermediate sizes were difficult to decipher, especially without a sonification legend. For the rhythmic node sonification, we found significant differences in mapping across paper clusters of different sizes. However, this was found only between two sizes. Results suggest that this interaction effect may be caused by the video stimuli. In transition sonification, we found significant pitch mapping effects but not timbral mapping effects, suggesting that this feature is harder to perceive. However, the main interaction effects were found between length and timbre. Our findings suggest that factors related to preferences influenced mapping choices.

7 Conclusion

This paper presents a novel multisensory interface for exploring bibliometric network data, enhancing traditional methods through combined sonification and visualization. Evaluation results provide a strong foundation for future development.

RQ1, explored through a think-aloud experiment, showed that the interface was more engaging than traditional graphs and offered new data perspectives, though further refinement is needed to improve user experience. **RQ2**, examined through the think-aloud test, showed that different interaction modes offered varied insights into the data, with user preferences depending on their interests. The animation prompted both usability- and affect-related responses. **RQ3**, explored through the online experiment, showed that established cross-modal mappings were consistently preferred in graph interaction modes.

The developed interface is a multimodal tool in which sound and visual design complement each other, improving interpretation of bibliometric data in networked graph representations. Future work will repeat the study with a higher number of participants and evaluate the mappings in an interactive setting. Also, the interface will be further enhanced with plugin support and additional data tools, aiming to enable more engaging data exploration and encourage the development of open-source multisensory interfaces.

8 Online Resources

The code for the interactive sonification interface can be accessed at a GitHub Repository[8]. The user testing documentation from Experiment 1, as well as the lab.js code, participants' data, and statistical analysis R code for Experiment 2, can be accessed online[9].

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